

# *Existence of $n$ -th root*

PROVING THE EXISTENCE OF  $n$ -TH ROOT is the topic of this note.

This note is an experiment in including PDF pages. The included PDF pages are scans of handwriting with a fountain pen. The proof is a modification of the proof of the existence of the square root from Chapter 2 of the *Lectures on Real Analysis* textbook<sup>1</sup> to the  $n$ -th root case.

<sup>1</sup> F. Lárusson. *Lectures on Real Analysis*. Australian Mathematical Society Lecture Series. Cambridge University Press, 2012. ISBN 9781107026780. URL <https://books.google.com/books?id=koj-IrXXwocC>

-1-

There is  $s \in \mathbb{R}$  with  $s^n = y$  for  $y > 0$  and  $n \geq 2$

Proof:

consider set  $A = \{x \in \mathbb{R} : x^n \leq y\}$

$$0 \in A \Rightarrow A \neq \emptyset$$

$\forall x \in A: x \leq y \Rightarrow y$  upper bound

$\Rightarrow s := \sup A$  exists

Claim:  $s^n = y$

We show this by eliminating the possibilities  $s^n < y, s^n > y$

Assume  $s^n < y$

We will work with expression  $(s+\varepsilon)^n$

$$(s+\varepsilon)^n = \sum_{k=0}^n \binom{n}{k} s^{n-k} \varepsilon^k$$

$$= s^n + \sum_{k=1}^n \binom{n}{k} s^{n-k} \varepsilon^k$$

$$= s^n + \varepsilon \cdot \sum_{k=1}^n \binom{n}{k} s^{n-k} \varepsilon^{k-1}$$

we will make sure that

$$0 < \varepsilon < 1$$

$$\Rightarrow (s+\varepsilon)^n < s^n + \varepsilon \underbrace{\sum_{k=1}^n \binom{n}{k} s^{n-k}}_{B:=}$$

-2-

Now choose  $\varepsilon$  such that

$$0 < \varepsilon < \min\left(1, \frac{y - s^n}{B}\right)$$

$$\Rightarrow (s + \varepsilon)^n < s^n + \varepsilon \cdot B < s^n + \frac{y - s^n}{B} \cdot B = y$$

$$\Rightarrow s + \varepsilon \in A \quad \swarrow \text{contradicts } s = \sup A$$

Assume  $s^n > y$ We will work with expression  $(s - \varepsilon)^n$ 

$$(s - \varepsilon)^n = \sum_{k=0}^n \binom{n}{k} s^{n-k} (-\varepsilon)^k$$

$$= s^n + \sum_{k=1}^n \binom{n}{k} s^{n-k} (-\varepsilon)^k$$

$$= s^n + \sum_{\substack{k=1 \\ k \equiv 0 \pmod{2}}}^n \binom{n}{k} s^{n-k} \varepsilon^k$$

$$- \sum_{\substack{k=1 \\ k \equiv 1 \pmod{2}}}^n \binom{n}{k} s^{n-k} \varepsilon^k$$

$$> s^n - \sum_{\substack{k=1 \\ k \equiv 1 \pmod{2}}}^n \binom{n}{k} s^{n-k} \varepsilon^k$$

$$= s^n - \varepsilon \cdot \sum_{\substack{k=1 \\ k \equiv 1 \pmod{2}}}^n \binom{n}{k} s^{n-k} \varepsilon^{k-1}$$

Again, we make

such  $0 < \varepsilon < 1$ 

$$\Rightarrow (s - \varepsilon)^n > s^n - \underbrace{\varepsilon \sum_{\substack{k=1 \\ k \equiv 1 \pmod{2}}}^n \binom{n}{k} s^{n-k}}_{C :=}$$

-3-

Now we choose  $\varepsilon$  such that

$$0 < \varepsilon < \min\left(1, \frac{s^n - y}{c}\right)$$

$$\Rightarrow (s - \varepsilon)^n > s^n - \varepsilon \cdot c > s^n - \frac{s^n - y}{c} \cdot c = y$$

 $\Rightarrow s - \varepsilon$  is upper bound of  $A$ 
 $\hookrightarrow$  contradicts  $s = \sup A$ 

$$\text{So } s^n = y$$

■

## *Bibliography*

F. Lárusson. *Lectures on Real Analysis*. Australian Mathematical Society Lecture Series. Cambridge University Press, 2012. ISBN 9781107026780. URL <https://books.google.com/books?id=koj-IrXXwocC>.